Implementation of A* Algorithm

PROGRAM :-

```
import heapq
def heuristic(a, g):
    return abs(a[0]-g[0])+abs(a[1]-g[1])
def astar(start, goal):
    open_list = [(heuristic(start, goal), 0, start, [start])]
    closed = set()
    while open_list:
        f, cost, node, path = heapq.heappop(open_list)
        if node == goal:
            return path
        if node in closed:
            continue
        closed.add(node)
        x, y = node
        for n in [(x+1, y), (x, y+1), (x, y-1), (x-1, y)]:
            if 0 <= n[0] <= 3 and 0 <= n[1] <= 3:
                if n not in closed:
                    g = cost + 1
                    heapq.heappush(open_list, (g+heuristic(n, goal),
                                               g, n, path + [n]))
print("Path:", astar((0,0), (3,3)))
```

output :-

Path: [(0,0), (1,0), (2,0), (3,0),
(3,1), (3,2), (3,3)]